Mathematical Statistics II

Lecture notes based on Kim, *Mathematical Statistics*, Ch. 6–11

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Chapter 6. Estimation

Setting

$X_1, \dots, X_n$ is a random sample from $f(x;\theta)$, $\theta \in \Omega \subseteq \mathbb{R}^k$. An **estimator** is a statistic $\hat{\theta}_n$ used to guess θ; it is **consistent** if $\hat{\theta}_n \xrightarrow{p} \theta$ for every $\theta \in \Omega$.

6.1 The method of moments

If $\eta = g(m_1, \dots, m_k)$ is a function of the population moments $m_r = E(X_1^r)$, the **method-of-moments estimator** (MME) replaces them by the sample moments $\hat{m}_r = \frac{1}{n}\sum X_i^r$:

$$\hat{\eta} = g(\hat{m}_1, \dots, \hat{m}_k).$$

Theorem 6.1

If $E|X_1|^k < \infty$ and g is continuous at $(m_1, \dots, m_k)$, the MME is consistent. If moreover $EX_1^{2k} < \infty$ and g is differentiable, it is asymptotically normal.

Proof. The vector of sample moments converges in probability to $(m_1, \dots, m_k)$ by the law of large numbers, and satisfies a multivariate CLT when $2k$ moments exist. Apply the continuous mapping theorem and the delta method (Chapter 5). ■

- $\hat{\sigma}^2_{\text{MME}} = \hat{m}_2 - \hat{m}_1^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$.
- The MME of a correlation, from bivariate sample moments, is the sample correlation r.
- MMEs are not unique: for Poisson(λ), $\lambda = m_1 = m_2 - m_1^2$ gives both $\bar{X}$ and $\frac{1}{n}\sum(X_i - \bar{X})^2$.

Example 6.1

For Gamma(α, β), $m_1 = \alpha\beta$ and $m_2 - m_1^2 = \alpha\beta^2$, so $\hat{\beta} = \hat{\sigma}^2/\bar{X}$ and $\hat{\alpha} = \bar{X}^2/\hat{\sigma}^2$. With $\bar{x} = 6$ and $\hat{\sigma}^2 = 12$: $\hat{\beta} = 2$, $\hat{\alpha} = 3$.

6.2 Maximum likelihood: one parameter

Definition (Likelihood and MLE)

The **likelihood** is $L(\theta) = \prod_i f(x_i;\theta)$, viewed as a function of θ for fixed data, and $l(\theta) = \log L(\theta)$. A **maximum likelihood estimator** (MLE) $\hat{\theta}$ satisfies $L(\hat{\theta}) = \max_{\theta\in\Omega} L(\theta)$.

The MLE chooses the parameter that makes the observed data most probable. When l is differentiable it solves the **likelihood equation** $\dot{l}(\theta) = 0$, but a root need not be the maximum.

Theorem 6.2

Let $\Omega = (a, b)$ and l be twice differentiable with $\ddot{l}(\theta) < 0$ on Ω and $l(\theta) \to -\infty$ at both ends. Then the likelihood equation has a unique root, and it is the MLE.

Proof. $\dot{l}$ is strictly decreasing, so it has at most one zero. Since $l \to -\infty$ at both ends and l is continuous, l attains a maximum inside Ω, where $\dot{l} = 0$. ■

Theorem 6.3 (Invariance)

If $\hat{\theta}$ is the MLE of θ and $\eta = g(\theta)$ is one-to-one, the MLE of η is $g(\hat{\theta})$. More generally, if g_1 is any function, define the MLE of $g_1(\theta)$ as $g_1(\hat{\theta})$; this agrees with the maximizer of the induced likelihood whenever g_1 can be completed to a one-to-one map (g_1, g_2).

Proof. The likelihood in terms of η is $L(g^{-1}(\eta))$, which is maximized where $g^{-1}(\eta) = \hat{\theta}$. For g_1, apply this to the one-to-one (g_1, g_2) and take the first coordinate. ■

Reparametrization can make the problem easier. For $\mathrm{Exp}(\theta)$ (mean θ), $l(\theta) = -n\log\theta - n\bar{x}/\theta$ is not concave, but in $\lambda = 1/\theta$, $l = n\log\lambda - n\bar{x}\lambda$ is strictly concave. So $\hat{\lambda} = 1/\bar{X}$ and $\hat{\theta} = \bar{X}$.

Exponential families. If $f(x;\eta) = \exp\{\eta T(x) - A(\eta) + S(x)\}$ with natural parameter η in an open interval, then $\dot{l}(\eta) = \sum T(x_i) - nA'(\eta)$ and $\ddot{l} = -nA''(\eta) = -n\,\mathrm{Var}_\eta T < 0$. Hence the MLE solves $E_\eta T(X) = \frac{1}{n}\sum T(x_i)$ (a moment equation) whenever a solution exists. Bernoulli, Poisson, exponential and normal (with one parameter known) are examples.

Example 6.2

For $U(0,\theta)$, $L(\theta) = \theta^{-n} I(\theta \geq x_{(n)})$ is decreasing on $[x_{(n)}, \infty)$, so $\hat{\theta} = X_{(n)}$. No likelihood equation is involved, and the support depends on θ; regularity conditions fail.

Numerical solution. When the equation has no closed form (e.g. logistic location), use **Newton–Raphson**: $\hat{\theta}^{(r+1)} = \hat{\theta}^{(r)} - \ddot{l}(\hat{\theta}^{(r)})^{-1}\dot{l}(\hat{\theta}^{(r)})$, started from a consistent estimate such as the MME.

6.3 Maximum likelihood: several parameters

For $\theta \in \mathbb{R}^k$ the MLE maximizes $l(\theta)$, typically by solving $\nabla l(\theta) = 0$. The multivariate analogue of Theorem 6.2: if Ω is open and convex, the Hessian $\nabla^2 l$ is negative definite on Ω, and $l \to -\infty$ at the boundary of Ω, then the root of $\nabla l = 0$ is unique and is the MLE. Invariance (Theorem 6.3) holds verbatim.

Example 6.3 (Normal)

For $N(\mu, \sigma^2)$, $\sum(x_i - \mu)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$, so l is maximized in μ at $\bar{x}$ for any σ^2. Then maximizing $-\frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum(x_i - \bar{x})^2$ gives $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$. By invariance, the MLE of $|\mu|$ is $|\bar{X}|$ and that of σ is $\hat{\sigma}$.

Example 6.4 (Multinomial)

For counts $x_1, \ldots, x_k$ with $\sum x_j = n$, $l = \sum_j x_j \log p_j$ on the simplex. Writing $\eta_j = \log(p_j/p_k)$ turns it into an exponential family with concave log-likelihood, and the MLE is $\hat{p}_j = x_j/n$ (when all $0 < x_j < n$; boundary cases follow by direct maximization).

In multiparameter exponential families $f = \exp\{\eta' T(x) - A(\eta) + S(x)\}$, again $\nabla^2 l = -n\,\mathrm{Var}_\eta T$ is negative definite, so the MLE solves $E_\eta T = \bar{T}$.

6.4 Asymptotic properties of the MLE

Definition (Fisher information)

For one observation, the **score** is $\dot\ell(\theta;X)=\frac{\partial}{\partial\theta}\log f(X;\theta)$ and the **Fisher information** is $I(\theta)=\operatorname{Var}_\theta\dot\ell(\theta;X)$. In matrix form, $I(\theta)=\operatorname{Var}_\theta\nabla\log f(X;\theta)$.

Lemma 6.4

Under regularity conditions (common support not depending on θ, and differentiation under the integral sign allowed): (a) $E_\theta\dot\ell(\theta;X)=0$; (b) $I(\theta)=-E_\theta\ddot\ell(\theta;X)$; (c) (Kullback–Leibler) $E_\theta\log f(X;\theta')<E_\theta\log f(X;\theta)$ for $\theta'\neq\theta$, if the model is identifiable.

Proof. (a) Differentiate $\int f(x;\theta)\,dx=1$: $\int\dot\ell f\,dx=0$. (b) Differentiate again: $\int(\ddot\ell+\dot\ell^2)f\,dx=0$. (c) By Jensen with the strictly concave log, $E_\theta\log\frac{f(X;\theta')}{f(X;\theta)}<\log E_\theta\frac{f(X;\theta')}{f(X;\theta)}=\log 1=0$, unless $f(\cdot;\theta')=f(\cdot;\theta)$ a.e. ■

Theorem 6.5 (Consistency)

Suppose each l_n is strictly concave with $l_n\to-\infty$ at the boundary of Ω (as in Theorem 6.2), and $\theta'\mapsto E_\theta\log f(X;\theta')$ is finite and continuous. Then the MLE is the unique root of the likelihood equation and $\hat\theta_n\xrightarrow{p}\theta$.

Proof (one parameter). For fixed $\epsilon>0$, the law of large numbers gives $\frac1n\{l_n(\theta\pm\epsilon)-l_n(\theta)\}\xrightarrow{p}E_\theta\log\frac{f(X;\theta\pm\epsilon)}{f(X;\theta)}<0$ by Lemma 6.4(c). So with probability $\to1$, $l_n(\theta)>l_n(\theta\pm\epsilon)$. By concavity the maximizer then lies in $(\theta-\epsilon,\theta+\epsilon)$. ■

Theorem 6.6 (Asymptotic normality)

Under regularity conditions (in addition to those above: $\log f$ three times differentiable in θ with $|\dddot\ell(\theta';x)|\le M(x)$ near θ, $E_\theta M(X)<\infty$, and $0<I(\theta)<\infty$),

$$\sqrt n(\hat\theta_n-\theta)\xrightarrow{d}N(0,I(\theta)^{-1}),\qquad\text{and in } k \text{ dimensions}\quad\sqrt n(\hat\theta_n-\theta)\xrightarrow{d}N_k(\mathbf0,I(\theta)^{-1}).$$

Proof. Expand the likelihood equation around θ:

$$0=\tfrac1n\dot l_n(\hat\theta_n)=\tfrac1n\dot l_n(\theta)+\tfrac1n\ddot l_n(\theta)(\hat\theta_n-\theta)+\tfrac12\tfrac1n\dddot l_n(\theta^*)(\hat\theta_n-\theta)^2.$$

By the CLT and Lemma 6.4(a), $\frac{1}{\sqrt n}\dot l_n(\theta)\xrightarrow{d}N(0,I(\theta))$. By the LLN and Lemma 6.4(b), $-\frac1n\ddot l_n(\theta)\xrightarrow{p}I(\theta)$. The remainder is bounded by $\frac12(\frac1n\sum M(X_i))|\hat\theta_n-\theta|\xrightarrow{p}0$ by consistency. Hence

$$\sqrt n(\hat\theta_n-\theta)=\big(I(\theta)+o_p(1)\big)^{-1}\tfrac{1}{\sqrt n}\dot l_n(\theta)\xrightarrow{d}N(0,I(\theta)^{-1})$$

by Slutsky. The vector proof is identical with gradients and Hessians. ■

So $\hat\theta_n\pm z_{\alpha/2}/\sqrt{nI(\hat\theta_n)}$ is an asymptotic interval, and by the delta method $\sqrt n(g(\hat\theta_n)-g(\theta))\xrightarrow{d}N(0,g'(\theta)^2/I(\theta))$.

Example 6.5

Bernoulli: $I(p)=\frac{1}{p(1-p)}$, giving $\sqrt n(\hat p-p)\xrightarrow{d}N(0,p(1-p))$. Poisson: $I(\lambda)=1/\lambda$. Exponential with mean θ: $I(\theta)=1/\theta^2$. Normal (μ,σ^2): $I=\operatorname{diag}(1/\sigma^2,1/(2\sigma^4))$, so $\hat\mu$ and $\hat\sigma^2$ are

asymptotically independent with variances σ^2/n and $2\sigma^4/n$.

Theorem 6.7 (One-step estimator)

If $\hat{\theta}_n^{(0)}$ is $\sqrt{n}$-consistent ($\sqrt{n}(\hat{\theta}_n^{(0)} - \theta)$ has a limiting distribution), then one Newton step

$$\hat{\theta}_n^{(1)} = \hat{\theta}_n^{(0)} - \ddot{l}_n(\hat{\theta}_n^{(0)})^{-1}\dot{l}_n(\hat{\theta}_n^{(0)})$$

has the same limit distribution $N(0, I(\theta)^{-1})$ as the MLE.

Proof. Expanding $\dot{l}_n(\hat{\theta}_n^{(0)})$ around θ as above, $\sqrt{n}(\hat{\theta}_n^{(1)} - \theta) = \sqrt{n}(\hat{\theta}_n^{(0)} - \theta)\{1 - \ddot{l}_n(\theta^*)/\ddot{l}_n(\hat{\theta}_n^{(0)})\} - \ddot{l}_n(\hat{\theta}_n^{(0)})^{-1}\dot{l}_n(\theta)$. The braces are $o_p(1)$, and the last term behaves as in Theorem 6.6. ■

E.g. for the logistic location model, start at $\bar{X}$ (which is $\sqrt{n}$-consistent); one step already gives an asymptotically efficient estimator.

6.5 Least squares

Definition (Linear model)

$\mathbf{Y} = X\beta + \mathbf{e}$, with X an $n \times (p+1)$ known matrix of rank $p+1$, $E\mathbf{e} = \mathbf{0}$ and $\operatorname{Var}\mathbf{e} = \sigma^2 I_n$. The **least squares estimator** minimizes $\|\mathbf{Y} - X\beta\|^2$.

Theorem 6.8

Let $\Pi = X(X'X)^{-1}X'$. Then (a) $\Pi = \Pi' = \Pi^2$ and $\Pi X = X$ (orthogonal projection onto the column space of X); (b) $\|\mathbf{Y} - X\beta\|^2 = \|\Pi\mathbf{Y} - X\beta\|^2 + \|(I - \Pi)\mathbf{Y}\|^2$, so $X\hat{\beta} = \Pi\mathbf{Y}$ and $\hat{\beta} = (X'X)^{-1}X'\mathbf{Y}$; (c) $E\hat{\beta} = \beta$, $\operatorname{Var}\hat{\beta} = \sigma^2(X'X)^{-1}$, and $\hat{\sigma}^2 = \|\mathbf{Y} - X\hat{\beta}\|^2/(n-p-1)$ is unbiased for σ^2.

Proof. (a) Direct computation. (b) Write $\mathbf{Y} - X\beta = (\Pi\mathbf{Y} - X\beta) + (I - \Pi)\mathbf{Y}$; the cross term vanishes since $\Pi'(I - \Pi) = 0$ and $X\beta = \Pi X\beta$. The first term is 0 iff $X\beta = \Pi\mathbf{Y}$, i.e. $\beta = (X'X)^{-1}X'\mathbf{Y}$. (c) $\hat{\beta} = \beta + (X'X)^{-1}X'\mathbf{e}$. And $E\|(I - \Pi)\mathbf{e}\|^2 = \sigma^2\operatorname{tr}(I - \Pi) = \sigma^2(n - p - 1)$. ■

Under normal errors, $\hat{\beta}$ is also the MLE of β; the MLE of σ^2 divides by n instead of $n - p - 1$. Simple linear regression is the case $p = 1$: $\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$.

Chapter 7. Hypothesis Testing

7.1 The logic of tests

A test of $H_0 : \theta \in \Omega_0$ against $H_1 : \theta \in \Omega_1$ ($\Omega_0 \cap \Omega_1 = \emptyset$) is a rule that rejects H_0 when the data fall in a **critical region** C. Equivalently, a test is a function $\phi(\mathbf{x}) \in [0,1]$ giving the probability of rejecting; $\phi = I_C$ for a non-randomized test.

- **Type I error:** rejecting a true H_0. **Type II error:** accepting a false H_0.
- The **power function** is $\pi(\theta) = E_\theta \phi(\mathbf{X}) = P_\theta(\text{reject})$.
- The **size** is $\sup_{\theta \in \Omega_0} \pi(\theta)$. A test has **level** α if its size is $\le \alpha$.

The two errors cannot both be made small for fixed n. The convention is to bound the type I error by α and then make the power on Ω_1 as large as possible, so rejecting H_0 is the strong conclusion.

Randomized tests. With discrete data an exact size α may require randomizing on the boundary: reject when $T > c$, and with probability γ when $T = c$, where $P_{\theta_0}(T > c) + \gamma P_{\theta_0}(T = c) = \alpha$.

Example 7.1

$X \sim B(10, p)$, $H_0 : p = \frac{1}{2}$ against $p > \frac{1}{2}$, $\alpha = 0.05$. Here $P(X \ge 9) = 11/1024 = 0.0107$ and $P(X = 8) = 45/1024 = 0.0439$. The randomized test rejects when $X \ge 9$, and with probability $\gamma = (0.05 - 0.0107)/0.0439 = 0.893$ when $X = 8$.

p-values. For a test rejecting when $T \ge c$, the p-value is $\sup_{\theta \in \Omega_0} P_\theta(T \ge t_{\text{obs}})$; rejecting when $p \le \alpha$ is a level-α test.

7.2 Likelihood ratio tests

Definition (Likelihood ratio test)

With $\Omega = \Omega_0 \cup \Omega_1$ and MLEs $\hat\theta^{\Omega}$, $\hat\theta^{\Omega_0}$ over Ω and Ω_0, the LRT rejects H_0 for large

$$\Lambda = \frac{\max_{\Omega} L(\theta)}{\max_{\Omega_0} L(\theta)}, \qquad \text{i.e.} \qquad 2\log\Lambda = 2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} \ge c,$$

with c chosen so that the size is α.

Since $\max_{\Omega} L = \max\{\max_{\Omega_0} L, \max_{\Omega_1} L\}$, a large ratio means the data fit some $\theta \in \Omega_1$ much better than any $\theta \in \Omega_0$. In practice one simplifies the critical region to an equivalent statistic with a known null distribution.

Example 7.2 (Normal mean, σ unknown)

For $N(\mu, \sigma^2)$ and $H_0 : \mu = \mu_0$, $\hat\sigma^2_\Omega = \frac{1}{n}\sum(x_i - \bar x)^2$ and $\hat\sigma^2_0 = \frac{1}{n}\sum(x_i - \mu_0)^2 = \hat\sigma^2_\Omega + (\bar x - \mu_0)^2$. So

$$2\log\Lambda = n\log\frac{\hat\sigma^2_0}{\hat\sigma^2_\Omega} = n\log\Big(1 + \frac{T^2}{n-1}\Big), \qquad T = \frac{\sqrt{n}(\bar x - \mu_0)}{s}.$$

This is increasing in $|T|$, so the LRT is the two-sided t-test: reject when $|T| \ge t_{\alpha/2}(n-1)$.

Example 7.3 (Normal variance)

For $H_0 : \sigma^2 = \sigma_0^2$ with μ unknown, with $u = \sum(x_i - \bar{x})^2/\sigma_0^2$ we get $2\log\Lambda = u - n - n\log(u/n)$. This is convex in u with minimum at $u = n$, so the LRT rejects when $u \le c_1$ or $u \ge c_2$. Under H_0, $u \sim \chi^2(n-1)$; equal tails $c_1 = \chi^2_{1-\alpha/2}(n-1)$, $c_2 = \chi^2_{\alpha/2}(n-1)$ give the usual approximation to the exact LRT.

Other standard LRTs reduce in the same way: two-sample t (equal variances), the F-test for two variances, and binomial or Poisson tests (randomized for exactness).

7.3 Large-sample approximations

Theorem 7.1 (Wilks, simple null)

Under the regularity conditions of Theorem 6.6 with $\Omega \subseteq \mathbb{R}^k$ open, if $H_0 : \theta = \theta_0$ is true,

$$2\{l(\hat{\theta}_n) - l(\theta_0)\} \xrightarrow{d} \chi^2(k).$$

Proof. Taylor-expand around $\hat{\theta}_n$, where $\nabla l(\hat{\theta}_n) = 0$:

$$l(\theta_0) = l(\hat{\theta}_n) + \tfrac{1}{2}(\theta_0 - \hat{\theta}_n)'\nabla^2 l(\theta^*)(\theta_0 - \hat{\theta}_n).$$

By consistency and the LLN, $-\frac{1}{n}\nabla^2 l(\theta^*) \xrightarrow{p} I(\theta_0)$. Hence $2\{l(\hat{\theta}_n) - l(\theta_0)\} = \sqrt{n}(\hat{\theta}_n - \theta_0)'\{I(\theta_0) + o_p(1)\}\sqrt{n}(\hat{\theta}_n - \theta_0)$. With $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} Z \sim N_k(\mathbf{0}, I^{-1})$, the limit is $Z'IZ \sim \chi^2(k)$ (quadratic forms of normal vectors). ∎

Theorem 7.2 (Wilks, composite null)

Partition $\theta = (\xi, \eta)$ with $\xi \in \mathbb{R}^{k_1}$, and test $H_0 : \xi = \xi_0$ (η unrestricted, $k_0 = k - k_1$ free parameters). Under H_0,

$$2\{l(\hat{\theta}_n^{\Omega}) - l(\hat{\theta}_n^{0})\} \xrightarrow{d} \chi^2(k_1),$$

and the **Wald** and **score** (Rao) statistics

$$W_n = n(\hat{\xi}_n - \xi_0)'\big[\{I(\hat{\theta}_n)^{-1}\}_{11}\big]^{-1}(\hat{\xi}_n - \xi_0), \qquad R_n = \tfrac{1}{n}\nabla l(\hat{\theta}_n^0)' I(\hat{\theta}_n^0)^{-1}\nabla l(\hat{\theta}_n^0)$$

have the same $\chi^2(k_1)$ limit, and all three differ by $o_p(1)$.

Sketch. Write $I = \begin{pmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{pmatrix}$ and $I_{11\cdot 2} = I_{11} - I_{12}I_{22}^{-1}I_{21}$. Expansions as in Theorem 7.1 give $2\{l(\hat{\theta}^{\Omega}) - l(\theta_0)\} \approx \mathbf{U}'I^{-1}\mathbf{U}$ and $2\{l(\hat{\theta}^{0}) - l(\theta_0)\} \approx \mathbf{U}_2'I_{22}^{-1}\mathbf{U}_2$, where $\mathbf{U} = n^{-1/2}\nabla l(\theta_0) \xrightarrow{d} N(\mathbf{0}, I)$. Their difference is $\mathbf{Z}'I_{11\cdot 2}^{-1}\mathbf{Z}$ with $\mathbf{Z} = \mathbf{U}_1 - I_{12}I_{22}^{-1}\mathbf{U}_2 \xrightarrow{d} N(\mathbf{0}, I_{11\cdot 2})$ (block-inverse formula), so it is $\chi^2(k_1)$. Since $\sqrt{n}(\hat{\xi} - \xi_0) \approx I_{11\cdot 2}^{-1}\mathbf{Z}$ and $\{I^{-1}\}_{11} = I_{11\cdot 2}^{-1}$, W_n has the same leading term. For R_n, $\nabla_\eta l(\hat{\theta}^0) = 0$ and $n^{-1/2}\nabla_\xi l(\hat{\theta}^0) \approx \mathbf{Z}$. ∎

A null of the form $g_1(\theta) = \cdots = g_{k_1}(\theta) = 0$ is handled by reparametrizing so that these functions become coordinates.

Application: multinomial goodness of fit. For Multi$(n, \mathbf{p})$ and $H_0 : \mathbf{p} = \mathbf{p}(\eta)$ with $\eta \in \mathbb{R}^{k_0}$ (and $k-1$ free cell probabilities under Ω), Wilks gives

$$G^2 = 2\sum_j x_j \log\frac{x_j}{n\hat{p}_j^0} \xrightarrow{d} \chi^2(k - 1 - k_0),$$

and the Pearson statistic $X^2 = \sum_j \frac{(x_j - n\hat{p}_j^0)^2}{n\hat{p}_j^0}$ is asymptotically equivalent. (Expand $x \log \frac{x}{e} \approx (x-e) + \frac{(x-e)^2}{2e}$ and note $\sum(x_j - e_j) = 0$.)

Example 7.4 (Independence in an $r \times c$ table)

$H_0 : p_{ij} = p_{i\cdot} p_{\cdot j}$, with MLEs $\hat{p}_{i\cdot} = x_{i\cdot}/n$ and $\hat{p}_{\cdot j} = x_{\cdot j}/n$. The degrees of freedom are $(rc-1)-(r-1)-(c-1) = (r-1)(c-1)$. For the 2×2 table with rows $(30, 20)$ and $(20, 30)$, all expected counts are 25, so $X^2 = 4 \cdot \frac{25}{25} = 4.0 > \chi^2_{0.05}(1) = 3.84$: reject independence at the 5% level.

Chapter 8. Comparing Estimators

8.1 Criteria for comparing estimators

For an estimator $\hat{\eta}$ of $\eta = \eta(\theta)$, the **mean squared error** is

$$\mathrm{MSE}(\hat{\eta}, \theta) = E_\theta(\hat{\eta} - \eta)^2 = \mathrm{Var}_\theta\, \hat{\eta} + \{\mathrm{bias}_\theta(\hat{\eta})\}^2.$$

No estimator minimizes the MSE uniformly in θ: the constant $\hat{\eta} \equiv \eta(\theta_0)$ has MSE zero at θ_0. So one restricts the class, most commonly to **unbiased** estimators ($E_\theta \hat{\eta} = \eta$ for all θ), and seeks the **uniformly minimum variance unbiased estimator** (UMVUE).

Example 8.1

For $N(\mu, \sigma^2)$, $S^2 = \frac{1}{n-1}\sum (X_i - \bar{X})^2$ is unbiased with MSE $= \frac{2\sigma^4}{n-1}$. The MLE $\hat{\sigma}^2 = \frac{n-1}{n} S^2$ has MSE $= \frac{(2n-1)\sigma^4}{n^2}$, which is smaller. Unbiasedness is a restriction, not a guarantee of small error.

8.2 Sufficient statistics

Definition (Sufficiency)

A statistic $Y = u(\mathbf{X})$ is **sufficient** for θ if the conditional distribution of $\mathbf{X}$ given $Y = y$ does not depend on θ.

Once Y is known, the rest of the sample carries no information about θ. For Bernoulli(θ), $P_\theta(\mathbf{X} = \mathbf{x} \mid \sum X_i = y) = \binom{n}{y}^{-1}$ whenever $\sum x_i = y$, so $\sum X_i$ is sufficient.

Theorem 8.1 (Factorization)

$Y = u(\mathbf{X})$ is sufficient iff $\prod_i f(x_i; \theta) = k_1(u(\mathbf{x}), \theta)\, k_2(\mathbf{x})$ for some functions k_1, k_2.

Proof (discrete case). If the factorization holds, then for $u(\mathbf{x}) = y$,

$$P_\theta(\mathbf{X} = \mathbf{x} \mid Y = y) = \frac{k_1(y, \theta) k_2(\mathbf{x})}{\sum_{\mathbf{z}: u(\mathbf{z}) = y} k_1(y, \theta) k_2(\mathbf{z})} = \frac{k_2(\mathbf{x})}{\sum_{u(\mathbf{z}) = y} k_2(\mathbf{z})},$$

which is free of θ. Conversely, if Y is sufficient, $P_\theta(\mathbf{X} = \mathbf{x}) = P(\mathbf{X} = \mathbf{x} \mid Y = u(\mathbf{x}))\, P_\theta(Y = u(\mathbf{x}))$, which is the factorization with $k_2(\mathbf{x}) = P(\mathbf{X} = \mathbf{x} \mid Y = u(\mathbf{x}))$. ■

Examples: $\sum X_i$ for Poisson and exponential; $(\sum X_i, \sum X_i^2)$ for $N(\mu, \sigma^2)$; $X_{(n)}$ for $U(0, \theta)$, since

$$\prod_i \theta^{-1} I(0 < x_i < \theta) = \theta^{-n} I(x_{(n)} < \theta)\, I(x_{(1)} > 0).$$

For a k-parameter exponential family $f(x; \theta) = \exp\{\sum_j \eta_j(\theta) T_j(x) - B(\theta) + S(x)\}$, the vector of sums $\sum_i T_j(X_i)$, $j = 1, \ldots, k$, is sufficient.

Theorem 8.2 (Rao–Blackwell)

Let Y be sufficient and $\hat{\eta}$ any unbiased estimator of η. Then $\hat{\eta}^{RB} = E(\hat{\eta} \mid Y)$ is a statistic, is unbiased, and $\mathrm{Var}_\theta\, \hat{\eta}^{RB} \le \mathrm{Var}_\theta\, \hat{\eta}$ for all θ.

Proof. By sufficiency the conditional expectation does not involve θ, so it is a statistic. Unbiasedness follows from the tower property. The variance decomposition gives $\operatorname{Var}\hat\eta = E\operatorname{Var}(\hat\eta \mid Y) + \operatorname{Var} E(\hat\eta \mid Y) \geq \operatorname{Var}\hat\eta^{RB}$. ■

So we need only look among functions of a sufficient statistic.

8.3 Complete sufficient statistics and the UMVUE

Definition (Completeness)

Y is **complete** if $E_\theta g(Y) = 0$ for all $\theta \in \Omega$ implies $P_\theta(g(Y) = 0) = 1$ for all θ.

Theorem 8.3 (Lehmann–Scheffé)

If Y is complete sufficient, then (a) any unbiased estimator of the form $\delta(Y)$ is the (essentially unique) UMVUE; (b) equivalently, $E(\hat\eta \mid Y)$ is the UMVUE for every unbiased $\hat\eta$.

Proof. Let $\delta_1(Y), \delta_2(Y)$ both be unbiased. Then $E_\theta\{\delta_1(Y) - \delta_2(Y)\} = 0$ for all θ, so $\delta_1 = \delta_2$ a.s. by completeness. Given any unbiased $\hat\eta$, Rao–Blackwell gives the unbiased function $E(\hat\eta \mid Y)$ of Y with no larger variance, and by uniqueness it equals $\delta(Y)$. So $\delta(Y)$ has variance at most that of every unbiased estimator. ■

- **Binomial:** for $Y \sim B(n,\theta)$, $\sum_y g(y)\binom{n}{y}\lambda^y = 0$ for all $\lambda = \theta/(1-\theta) > 0$ forces every coefficient to vanish (a polynomial identity). So Y is complete and $\bar X$ is the UMVUE of θ.
- **Poisson:** the same argument with a power series in $\lambda = n\theta$.
- **Exponential:** $Y = \sum X_i \sim \text{Gamma}(n,\theta)$, and uniqueness of the Laplace transform gives completeness.
- **Exponential families:** if the natural parameter space contains a k-dimensional open set, $(\sum_i T_1, \ldots, \sum_i T_k)$ is complete. Hence $\bar X$ and S^2 are UMVUEs for a normal sample.

Example 8.2

For Poisson(θ) and $\eta = P(X = 0) = e^{-\theta}$, the indicator $I(X_1 = 0)$ is unbiased. Given $Y = \sum X_i = y$, $X_1 \sim B(y, 1/n)$, so the UMVUE is $E\{I(X_1 = 0) \mid Y\} = (1 - \frac{1}{n})^Y$. (The MLE $e^{-\bar X}$ is biased.)

Example 8.3

For $U(0,\theta)$, $Y = X_{(n)}$ has density ny^{n-1}/θ^n. If $\int_0^\theta g(y)y^{n-1}\,dy = 0$ for all θ, differentiating gives $g \equiv 0$, so Y is complete. Since $EY = \frac{n}{n+1}\theta$, the UMVUE is $\frac{n+1}{n}X_{(n)}$.

The information inequality.

Theorem 8.4 (Cramér–Rao)

Under the regularity conditions of Lemma 6.4, if $\hat\eta$ is unbiased for $\eta(\theta)$, with η differentiable, then $\operatorname{Var}_\theta \hat\eta \geq \dfrac{\eta'(\theta)^2}{nI(\theta)}$.

Proof. Differentiate $\int \hat\eta(\mathbf{x}) \prod f(x_i;\theta)\,d\mathbf{x} = \eta(\theta)$ under the integral sign: $\operatorname{Cov}_\theta(\hat\eta, \dot l_n(\theta)) = \eta'(\theta)$, since $E\dot l_n = 0$. By Cauchy–Schwarz, $\eta'(\theta)^2 \le \operatorname{Var}\hat\eta \cdot \operatorname{Var}\dot l_n = \operatorname{Var}\hat\eta \cdot nI(\theta)$. ∎

Equality holds iff $\dot l_n(\theta)$ is linear in $\hat\eta$, i.e. essentially only in exponential families with $\hat\eta$ the mean of the natural statistic. The bound can be unattainable even when a UMVUE exists, as in Example 8.2.

8.4 Asymptotic efficiency

Definition (Asymptotic relative efficiency)

If $\sqrt{n}(\hat\theta_n^{(i)} - \theta) \xrightarrow{d} N(0, \sigma_i^2(\theta))$ for $i = 1, 2$, then $ARE(\hat\theta^{(1)}, \hat\theta^{(2)}) = \sigma_2^2(\theta)/\sigma_1^2(\theta)$.

The ARE is the limiting ratio of sample sizes needed for the same precision: estimator 2 needs ARE times as many observations as estimator 1 for intervals of the same length. By Theorem 6.6 the MLE attains asymptotic variance $1/I(\theta)$. Under regularity no reasonable estimator does better, so the MLE is **asymptotically efficient**.

Example 8.4 (Location models)

For $f(x-\theta)$ with f symmetric, the mean has asymptotic variance σ^2, the median $1/(4f(0)^2)$ (Theorem 5.8 in Volume I), and the MLE $1/I$. With the standard logistic density, $\sigma^2 = \pi^2/3$, $f(0) = \tfrac14$ and $I = \tfrac13$. So the asymptotic variances are 3 (MLE), 3.29 (mean) and 4 (median), and $ARE(\text{mean}, \text{median}) = 12/\pi^2 \approx 1.22$. For the normal, $ARE(\text{mean}, \text{median}) = \pi/2$; for the Laplace, the median is the MLE and beats the mean by a factor of 2.

Chapter 9. Comparing Tests

9.1 Power and the choice of a test

Among level-α tests of $H_0 : \theta \in \Omega_0$ against $H_1 : \theta \in \Omega_1$, a test ϕ^* is **uniformly most powerful** (UMP) if $E_\theta \phi^* \geq E_\theta \phi$ for every $\theta \in \Omega_1$ and every level-α test ϕ.

9.2 Most powerful tests

Theorem 9.1 (Neyman–Pearson lemma)

For simple hypotheses $H_0 : \theta = \theta_0$ and $H_1 : \theta = \theta_1$ with joint densities f_0, f_1, the test

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & f_1(\mathbf{x}) > k f_0(\mathbf{x}), \\ \gamma, & f_1(\mathbf{x}) = k f_0(\mathbf{x}), \\ 0, & f_1(\mathbf{x}) < k f_0(\mathbf{x}), \end{cases}$$

with $k \geq 0$ and $\gamma \in [0, 1]$ chosen so that $E_0 \phi^* = \alpha$, is most powerful at level α.

Proof. For any level-α test ϕ, $(\phi^* - \phi)(f_1 - k f_0) \geq 0$ pointwise: where $f_1 > k f_0$, $\phi^* = 1 \geq \phi$; where $f_1 < k f_0$, $\phi^* = 0 \leq \phi$. Integrating, $E_1 \phi^* - E_1 \phi \geq k(E_0 \phi^* - E_0 \phi) = k(\alpha - E_0 \phi) \geq 0$. ∎

So the most powerful test rejects for large likelihood ratios f_1 / f_0. Randomization at the boundary is needed only for discrete data.

Example 9.1

$N(\mu, 1)$, $H_0 : \mu = 0$ against $\mu = 1$: $f_1 / f_0 = \exp(n\bar{x} - n/2)$ is increasing in $\bar{x}$, so the MP test rejects when $\bar{X} > z_\alpha / \sqrt{n}$. It does not depend on the value 1, so it is UMP against $\mu > 0$.

Theorem 9.2 (One-parameter exponential families)

If $f(x; \theta) = \exp\{g(\theta) T(x) - B(\theta) + S(x)\}$ with g increasing, the test that rejects when $\sum T(x_i) > c$ (randomizing at $= c$), with size α at θ_0, is UMP for $H_0 : \theta \leq \theta_0$ against $H_1 : \theta > \theta_0$.

Proof. For any $\theta_1 > \theta_0$, the ratio $f(\mathbf{x}; \theta_1) / f(\mathbf{x}; \theta_0)$ is increasing in $\sum T$. So ϕ^* is the MP test of θ_0 against θ_1, for every $\theta_1 > \theta_0$. Its power is nondecreasing in θ: for $\theta' < \theta''$, ϕ^* is MP of size $\alpha' = E_{\theta'} \phi^*$ for θ' against θ'', hence at least as powerful as the trivial test $\phi \equiv \alpha'$. Therefore $\sup_{\theta \leq \theta_0} E_\theta \phi^* = \alpha$, and enlarging H_0 to $\theta \leq \theta_0$ keeps the level. ∎

The same holds whenever the family has a **monotone likelihood ratio** in a statistic T. For two-sided alternatives no UMP test exists in general; one restricts to unbiased tests (power $\geq \alpha$ on H_1), and the equal-tail tests are then UMP unbiased in exponential families.

9.3 Large-sample power and efficiency of tests

Theorem 9.3

Let a level-α test of $H_0 : \theta = \theta_0$ against $\theta > \theta_0$ reject when $\sqrt{n}\{T_n - \mu(\theta_0)\}/\sigma(\theta_0) \geq z_\alpha$, where $\sqrt{n}\{T_n - \mu(\theta)\}/\sigma(\theta) \xrightarrow{d} N(0,1)$ locally uniformly. For local alternatives $\theta_n = \theta_0 + K/\sqrt{n}$,

$$\gamma_n(\theta_n) \to 1 - \Phi\Big(z_\alpha - K\frac{\dot\mu(\theta_0)}{\sigma(\theta_0)}\Big),$$

and the sample size needed for power $1-\beta$ at θ_1 is approximately

$$N \approx \Big(\frac{\sigma(\theta_0)}{\dot\mu(\theta_0)}\Big)^2 \Big(\frac{z_\alpha + z_\beta}{\theta_1 - \theta_0}\Big)^2.$$

Proof. Under θ_n, $\sqrt{n}\{T_n - \mu(\theta_0)\}/\sigma(\theta_0) = \sqrt{n}\{T_n - \mu(\theta_n)\}/\sigma(\theta_0) + \sqrt{n}\{\mu(\theta_n) - \mu(\theta_0)\}/\sigma(\theta_0)$. The first term is approximately $N(0,1)$, since $\sigma(\theta_n) \to \sigma(\theta_0)$. The second tends to $K\dot\mu(\theta_0)/\sigma(\theta_0)$, which gives the power limit. For the sample size, set $1 - \Phi(z_\alpha - \sqrt{N}(\theta_1-\theta_0)\dot\mu/\sigma) = 1-\beta$, i.e. $z_\alpha - \sqrt{N}(\theta_1-\theta_0)\dot\mu/\sigma = -z_\beta$, and solve for N. ∎

The quantity $e = (\dot\mu(\theta_0)/\sigma(\theta_0))^2$ is the **efficacy** of the test. The **Pitman ARE** of test 1 relative to test 2 is the limiting inverse ratio of sample sizes, $N_2/N_1 \to e_1/e_2$.

9.4 Nonparametric tests and their efficiency

In the location model $f(x-\theta)$ with f symmetric about 0, consider $H_0 : \theta = 0$ against $\theta > 0$.

- **Sign test:** $S = \sum I(X_i > 0) \sim B(n, \tfrac12)$ under H_0 for every continuous f (distribution-free). With $T_n = S/n$, $\mu(\theta) = 1 - F(-\theta)$, $\dot\mu(0) = f(0)$ and $\sigma(0) = \tfrac12$, so the efficacy is $4f(0)^2$.
- **Wilcoxon signed-rank test:** $W^+ = \sum_i R_i^+ I(X_i > 0)$, where R_i^+ is the rank of $|X_i|$. Under H_0 its distribution is that of $\sum_k k B_k$ with B_k i.i.d. Bernoulli($\tfrac12$), with mean $\frac{n(n+1)}{4}$ and variance $\frac{n(n+1)(2n+1)}{24}$. Its efficacy is $12(\int f^2)^2$.
- **t-test** (or z-test): efficacy $1/\sigma^2$.

Proposition 9.4

$ARE(\text{sign}, t) = 4\sigma^2 f(0)^2$, $ARE(W^+, t) = 12\sigma^2(\int f^2)^2$. For the normal these are $2/\pi \approx 0.64$ and $3/\pi \approx 0.955$; for the Laplace, 2 and 1.5; and $ARE(W^+, t) \geq 0.864$ for every symmetric f.

Proof. Take ratios of efficacies (Theorem 9.3). For the signed-rank statistic, $W^+/\binom{n+1}{2}$ is asymptotically equivalent to the U-statistic estimating $P(X_1 + X_2 > 0)$. Its derivative at $\theta = 0$ is $2\int f^2$, and its null standard deviation is $1/\sqrt{3n}$, which gives efficacy $12(\int f^2)^2$. The normal and Laplace values follow by direct substitution. The lower bound $0.864 = 108/125$ is attained by a parabolic density (Hodges–Lehmann). ∎

So distribution-free tests lose little under normality and can gain a lot under heavy tails. The two-sample analogues (Wilcoxon rank-sum versus two-sample t) have the same AREs.

Chapter 10. Analysis of Variance and Regression

10.1 One-way analysis of variance

Definition (One-way normal model)

$X_{ij} = \mu_i + e_{ij}$ with $e_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2)$, $i = 1, \dots, k$, $j = 1, \dots, n_i$, $n = \sum n_i$. Writing $\bar{\mu} = \sum n_i \mu_i / n$ and $\alpha_i = \mu_i - \bar{\mu}$ (so $\sum n_i \alpha_i = 0$), the model becomes $X_{ij} = \bar{\mu} + \alpha_i + e_{ij}$.

Theorem 10.1

(a) $(\bar{X}_1, \dots, \bar{X}_k, \sum_{i,j} (X_{ij} - \bar{X}_i)^2)$ is complete sufficient, so $\hat{\mu}_i = \bar{X}_i$, $\hat{\alpha}_i = \bar{X}_i - \bar{X}$ and $\hat{\sigma}^2 = SSE/(n-k)$ are UMVUEs, where $SSE = \sum_{i,j} (X_{ij} - \bar{X}_i)^2$. (b) The $\bar{X}_i \sim N(\mu_i, \sigma^2/n_i)$ are independent and independent of SSE, and $SSE/\sigma^2 \sim \chi^2(n-k)$.

Proof. (a) The joint density is a $(k+1)$-parameter exponential family in these statistics; apply Lehmann–Scheffé (Theorem 8.3). (b) Apply Theorem 4.2 of Volume I to each group and use independence across groups; chi-squares add. ∎

Decomposition and the F-test. With $SST = \sum_{i,j} (X_{ij} - \bar{X})^2$ and $SSTr = \sum_i n_i (\bar{X}_i - \bar{X})^2$,

$$SST = SSTr + SSE,$$

since the cross term $\sum_i (\bar{X}_i - \bar{X}) \sum_j (X_{ij} - \bar{X}_i)$ vanishes.

Theorem 10.2

The likelihood ratio test of $H_0 : \alpha_1 = \cdots = \alpha_k = 0$ rejects when

$$F = \frac{SSTr/(k-1)}{SSE/(n-k)} \geq F_\alpha(k-1, n-k),$$

and under H_0, $F \sim F(k-1, n-k)$.

Proof. The restricted and full MLEs of σ^2 are SST/n and SSE/n, so $2 \log \Lambda = n \log(SST/SSE) = n \log\left(1 + \frac{k-1}{n-k} F\right)$, which is increasing in F. Under H_0, $SSTr/\sigma^2 = \|(P_G - P_1)\mathbf{e}\|^2/\sigma^2$, where P_G projects onto group-mean vectors and P_1 onto constants. $P_G - P_1$ is idempotent of rank $k-1$ and orthogonal to $I - P_G$, so Theorem 4.9 of Volume I gives independent $\chi^2(k-1)$ and $\chi^2(n-k)$. ∎

Simultaneous intervals. For contrasts $\sum c_i \mu_i$ with $\sum c_i = 0$:

- **Scheffé**: $\sum c_i \bar{X}_i \pm \sqrt{(k-1) F_\alpha(k-1, n-k)}\, \hat{\sigma} \sqrt{\sum c_i^2 / n_i}$ holds simultaneously for *all* contrasts with confidence $1 - \alpha$.
- **Bonferroni**: for a fixed list of m comparisons, use $t_{\alpha/(2m)}(n-k)$ in each; this is shorter when m is small.

Why Scheffé works. By Cauchy–Schwarz, $\left(\sum_i c_i (\bar{X}_i - \mu_i)\right)^2 \leq \sum_i \frac{c_i^2}{n_i} \cdot \sum_i n_i \{(\bar{X}_i - \bar{X}) - (\mu_i - \bar{\mu})\}^2$ for contrasts (subtracting the common $\bar{X} - \bar{\mu}$ does not change a contrast). The last sum is $\sigma^2 \chi^2(k-1)$, independent of $\hat{\sigma}^2$, so the bound holds for all $\mathbf{c}$ simultaneously with probability $P(F(k-1, n-k) \leq F_\alpha) = 1 - \alpha$. ∎

10.2 Two-way analysis of variance

With r replications per cell, $X_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}$, $i = 1, \dots, a$, $j = 1, \dots, b$, $k = 1, \dots, r$, with the side conditions $\sum_i \alpha_i = \sum_j \beta_j = \sum_i \gamma_{ij} = \sum_j \gamma_{ij} = 0$. The orthogonal decomposition

$$SST = SSA + SSB + SSAB + SSE$$

uses $SSA = br \sum_i (\bar{X}_{i\cdot\cdot} - \bar{X})^2$, $SSB = ar \sum_j (\bar{X}_{\cdot j\cdot} - \bar{X})^2$, $SSAB = r \sum_{i,j} (\bar{X}_{ij\cdot} - \bar{X}_{i\cdot\cdot} - \bar{X}_{\cdot j\cdot} + \bar{X})^2$ and $SSE = \sum (X_{ijk} - \bar{X}_{ij\cdot})^2$. The degrees of freedom are $a-1$, $b-1$, $(a-1)(b-1)$ and $ab(r-1)$. Each null hypothesis (no A effect, no B effect, no interaction) is tested by the corresponding mean square over $SSE/\{ab(r-1)\}$, which has an F distribution under that null. The proof is the same projection argument: the four pieces are squared lengths of projections onto mutually orthogonal subspaces. With $r = 1$ there is no SSE; assume $\gamma_{ij} = 0$ and use $SSAB$ as the error.

10.3 Quadratic forms of normal vectors

Theorem 10.3 (Cochran)

Let $\mathbf{Z} \sim N_n(\mathbf{0}, I)$ and $I = A_1 + \cdots + A_m$ with symmetric A_j of ranks r_j. If $\sum r_j = n$, then the $\mathbf{Z}' A_j \mathbf{Z}$ are independent with $\chi^2(r_j)$ distributions.

Proof. $\sum r_j = n$ forces each A_j to be idempotent and $A_j A_l = 0$ for $j \neq l$. (Write each A_j as a sum of r_j rank-one terms $\pm \mathbf{v}\mathbf{v}'$; the n vectors together form a matrix V with $V \,\mathrm{diag}(\pm 1) V' = I$, which forces all signs $+$ and V orthogonal.) Then Theorem 4.9 of Volume I applies. ∎

If $\mathbf{Y} \sim N_n(\mu, \sigma^2 I)$ and A is a projection of rank r, then $\mathbf{Y}' A \mathbf{Y}/\sigma^2$ has a **noncentral** $\chi^2(r, \lambda)$ distribution with $\lambda = \mu' A \mu / \sigma^2$. Its mean is $r + \lambda$, which is why F tests have power increasing in λ.

10.4 Linear regression

In $\mathbf{Y} = X\beta + \mathbf{e}$ with $\mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I)$ and $\operatorname{rank} X = p + 1$:

Theorem 10.4

(a) $\hat{\beta} = (X'X)^{-1} X'\mathbf{Y} \sim N(\beta, \sigma^2 (X'X)^{-1})$; (b) $SSE = \|\mathbf{Y} - X\hat{\beta}\|^2$ satisfies $SSE/\sigma^2 \sim \chi^2(n - p - 1)$ and is independent of $\hat{\beta}$; (c) $\hat{\beta}$ and $\hat{\sigma}^2 = SSE/(n-p-1)$ are UMVUEs (Gauss–Markov: even without normality, $\mathbf{c}'\hat{\beta}$ has the smallest variance among linear unbiased estimators of $\mathbf{c}'\beta$).

Proof. (a) $\hat{\beta}$ is linear in a normal vector. (b) $SSE = \mathbf{e}'(I - \Pi)\mathbf{e}$, a projection of rank $n - p - 1$. $\hat{\beta} - \beta = (X'X)^{-1}X'\mathbf{e}$ and $(X'X)^{-1}X'(I - \Pi) = 0$, so they are independent. (c) Completeness of $(X'\mathbf{Y}, \mathbf{Y}'\mathbf{Y})$ in the exponential family. For Gauss–Markov, any linear unbiased $\mathbf{a}'\mathbf{Y}$ has $X'\mathbf{a} = \mathbf{c}$; write $\mathbf{a} = X(X'X)^{-1}\mathbf{c} + \mathbf{d}$ with $X'\mathbf{d} = 0$. Then $\operatorname{Var}(\mathbf{a}'\mathbf{Y}) = \sigma^2(\mathbf{c}'(X'X)^{-1}\mathbf{c} + \|\mathbf{d}\|^2)$. ∎

Consequently $(\hat{\beta}_j - \beta_j)/\{\hat{\sigma}\sqrt{(X'X)^{-1}_{jj}}\} \sim t(n-p-1)$. For a submodel with q fewer columns,

$$F = \frac{(SSE_0 - SSE)/q}{SSE/(n-p-1)} \sim F(q, n-p-1)$$

under the null, which is the LRT. At a new point $\mathbf{x}_0$, the interval for the mean response is $\mathbf{x}_0'\hat{\beta} \pm t_{\alpha/2}\hat{\sigma}\sqrt{\mathbf{x}_0'(X'X)^{-1}\mathbf{x}_0}$. The prediction interval for a new observation adds 1 under the root.

The one-way model fits the same framework, with an $n \times (k+1)$ design matrix of rank k made identifiable by the constraint $\sum n_i \alpha_i = 0$.

Chapter 11. Bayesian Inference

11.1 Prior and posterior distributions

In Bayesian inference the parameter is random: a **prior** density $\pi(\theta)$ expresses belief before the data, and the model $f(\mathbf{x} \mid \theta)$ is the conditional law of the data. Bayes' theorem gives the **posterior**

$$\pi(\theta \mid \mathbf{x}) = \frac{f(\mathbf{x} \mid \theta)\pi(\theta)}{\int f(\mathbf{x} \mid \theta')\pi(\theta')\,d\theta'} \;\propto\; L(\theta; \mathbf{x})\,\pi(\theta).$$

A family of priors is **conjugate** if the posterior stays in it.

Proposition 11.1 (Conjugate pairs)

(a) Bernoulli(θ) data with Beta(α_1, α_2) prior: the posterior is Beta($\alpha_1 + t, \alpha_2 + n - t$), $t = \sum x_i$.

(b) Poisson(θ) data with Gamma(α, β) prior: the posterior is Gamma($\alpha + t, \beta/(1 + n\beta)$).

(c) $N(\theta, \sigma^2)$ data (σ^2 known) with $N(\mu_0, \tau^2)$ prior: the posterior is $N(\mu_n, \tau_n^2)$ with

$$\frac{1}{\tau_n^2} = \frac{n}{\sigma^2} + \frac{1}{\tau^2}, \qquad \mu_n = \tau_n^2\Big(\frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\tau^2}\Big).$$

Proof. Multiply likelihood and prior and recognize the kernel. For (c), $-\frac{n(\bar{x}-\theta)^2}{2\sigma^2} - \frac{(\theta-\mu_0)^2}{2\tau^2}$ is quadratic in θ with leading coefficient $-\frac{1}{2\tau_n^2}$; complete the square. ∎

In (c) the posterior mean is a precision-weighted average of $\bar{x}$ and μ_0, and the posterior precision is the sum of the data and prior precisions. As $n \to \infty$ the prior's influence vanishes.

Normal model with both parameters unknown. With the normal–inverse-gamma prior $\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2/\kappa_0)$ and $\nu_0\sigma_0^2/\sigma^2 \sim \chi^2(\nu_0)$, the posterior has the same form with $\kappa_n = \kappa_0 + n$, $\mu_n = (\kappa_0\mu_0 + n\bar{x})/\kappa_n$, $\nu_n = \nu_0 + n$ and $\nu_n\sigma_n^2 = \nu_0\sigma_0^2 + (n-1)s^2 + \frac{\kappa_0 n}{\kappa_n}(\bar{x} - \mu_0)^2$. Integrating out σ^2, the marginal posterior is $\frac{\mu - \mu_n}{\sigma_n/\sqrt{\kappa_n}} \sim t(\nu_n)$.

11.2 Bayesian inference

All inference uses the posterior.

- **Point estimates:** the posterior mean, median or mode.
- **Credible interval:** a set C with $P(\theta \in C \mid \mathbf{x}) = 1 - \alpha$, e.g. equal-tailed posterior quantiles. It is a probability statement about θ given the data, unlike a confidence interval.
- **Testing:** compare $P(\Omega_0 \mid \mathbf{x})$ with $P(\Omega_1 \mid \mathbf{x})$, or use the Bayes factor, the ratio of posterior odds to prior odds.

Example 11.1

With a uniform Beta(1, 1) prior and $t = 7$ successes in $n = 10$ trials, the posterior is Beta(8, 4), with mean $8/12 = 0.667$ and mode $7/10$ (the MLE). In the normal–normal case with $n = 4$, $\sigma = 2$, $\bar{x} = 3$ and prior $N(0, 1)$: the posterior precision is $4/4 + 1 = 2$, so $\tau_n^2 = \frac{1}{2}$ and $\mu_n = \frac{1}{2}(3 + 0) = 1.5$. The 95% credible interval is $1.5 \pm 1.96\sqrt{0.5} = (0.11, 2.89)$.

For large n, under regularity, the posterior is approximately $N(\hat{\theta}_{\text{MLE}}, \{nI(\hat{\theta})\}^{-1})$ (Bernstein–von Mises). So credible and confidence intervals agree asymptotically.

11.3 Bayes and minimax estimators

With a loss $L(\theta, \hat{\eta})$, the **risk** is $R_{\hat{\eta}}(\theta) = E_\theta L(\theta, \hat{\eta}(\mathbf{X}))$, and the **Bayes risk** is $r(\pi, \hat{\eta}) = \int R_{\hat{\eta}}(\theta)\pi(\theta)\, d\theta$. A **Bayes estimator** minimizes the Bayes risk. A **minimax estimator** minimizes $\sup_\theta R_{\hat{\eta}}(\theta)$.

Theorem 11.2

The Bayes estimator minimizes the posterior expected loss $E\{L(\theta, a) \mid \mathbf{x}\}$ over a, for each $\mathbf{x}$. For squared error it is the posterior mean; for absolute error, the posterior median; for weighted squared error $w(\theta)(a-\eta)^2$, it is $E\{w\eta \mid \mathbf{x}\}/E\{w \mid \mathbf{x}\}$.

Proof. $r(\pi, \hat{\eta}) = \int \Big[\int L(\theta, \hat{\eta}(\mathbf{x}))\pi(\theta \mid \mathbf{x})\, d\theta \Big] m(\mathbf{x})\, d\mathbf{x}$ with m the marginal density of $\mathbf{X}$, so it is minimized pointwise in $\mathbf{x}$. For squared error, $E\{(a-\theta)^2 \mid \mathbf{x}\}$ is minimized at $a = E(\theta \mid \mathbf{x})$; for absolute error, at a median; the weighted case follows by differentiating in a. ■

For instance, under the loss $(\hat{\sigma}^2/\sigma^2 - 1)^2$ in the normal model, the Bayes estimator is $E(\sigma^{-2} \mid \mathbf{x})/E(\sigma^{-4} \mid \mathbf{x}) = \frac{\nu_n}{\nu_n+2}\sigma_n^2$, the posterior mode.

Theorem 11.3

If $\hat{\eta}^*$ is Bayes for some prior π and has constant risk, then $\hat{\eta}^*$ is minimax.

Proof. For any $\hat{\eta}$: $\sup_\theta R_{\hat{\eta}} \geq r(\pi, \hat{\eta}) \geq r(\pi, \hat{\eta}^*) = R_{\hat{\eta}^*}(\theta_0) = \sup_\theta R_{\hat{\eta}^*}$. ■

Example 11.2 (Binomial minimax)

For Beta(α_1, α_2) priors the Bayes estimator under squared error is $\frac{t+\alpha_1}{n+\alpha_1+\alpha_2}$, of the form $a\hat{\theta} + b$ with $\hat{\theta} = t/n$. Its risk is $a^2\theta(1-\theta)/n + (b + (a-1)\theta)^2$, which is constant in θ iff $\alpha_1 = \alpha_2 = \sqrt{n}/2$. Hence

$$\hat{\theta}^* = \frac{T + \sqrt{n}/2}{n + \sqrt{n}}, \qquad R(\theta) \equiv \frac{1}{4(1+\sqrt{n})^2},$$

is minimax. Its risk is below $\theta(1-\theta)/n$ only near $\theta = \frac{1}{2}$, so the minimax estimator is not uniformly better than $\bar{X}$.